

Measure of accuracy, Bias, Confounder & chance

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Measurements

- **Measurement**

- quantity that has both a number and a unit
- fundamental to experimental science so must be able to take and evaluate them



Evaluating measurements

- **Accuracy**
 - measure of how close a measurement comes to the actual or true value of whatever is measured.
- **Precision**
 - measure of how close a series of measurements are to one another.



Evaluating measurements

- **Accuracy vs. Precision**



a
Good accuracy
Good precision



b
Poor accuracy
Good precision



c
Poor accuracy
Poor precision



Evaluating measurements

Precision

- ❖ reproducibility
- ❖ check by repeating measurements
- ❖ poor precision results from poor technique.

Accuracy

- ❖ correctness
- ❖ check by using a different method
- ❖ poor accuracy results from procedural or equipment flaws.



Evaluating measurements

- Often, we are experimentally determining a value in the lab that is already known.
 - When we do this, we must calculate error in order to see how **accurate and precise** our results are
- In lab reports, you will be required to determine your error and percent error.



Evaluating measurements

- To determine **Error**:
 - The **accepted value** is the correct value based on reliable references.
 - The **experimental value** is the value measured in the lab.
 - The difference between the experimental value and the accepted value is called the **error**.

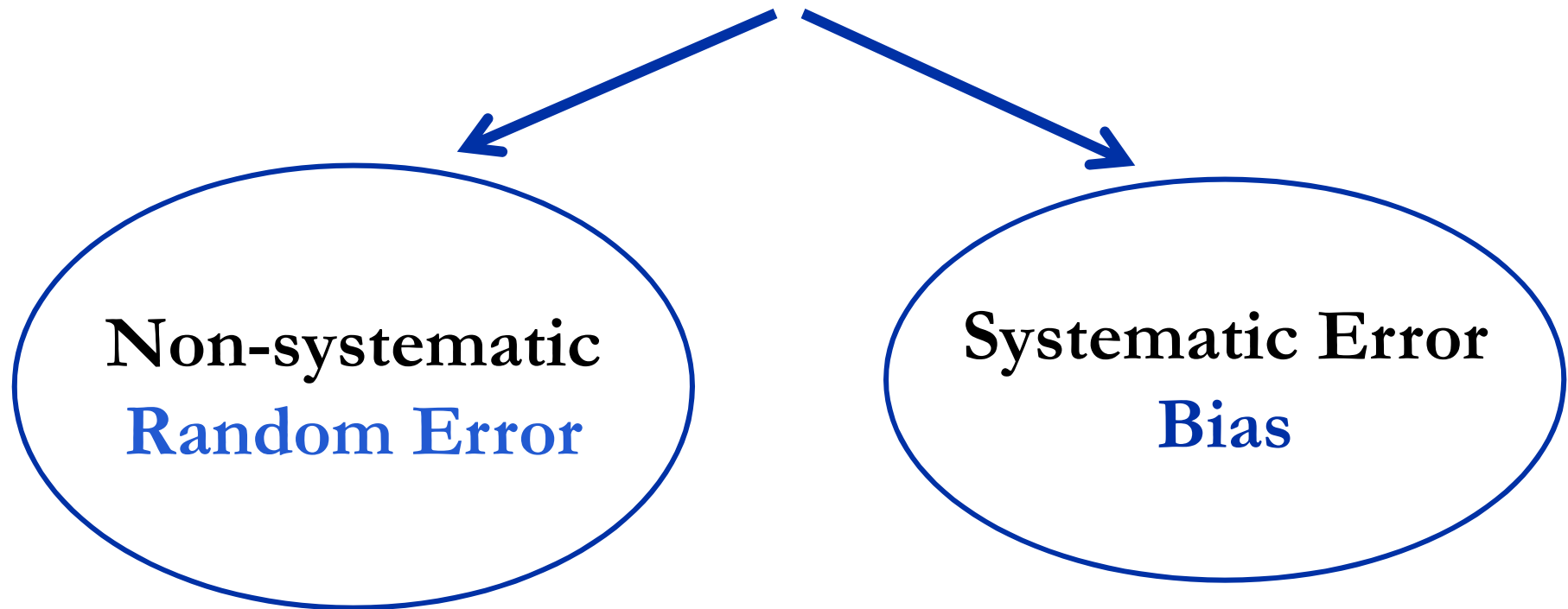
$$\text{Error} = \text{experimental value} - \text{accepted value}$$



Source of variation in Measurements

- **During observations**
 - by a single person (intra-observer variation) or
 - by different people (inter-observer variation).
 - Eg. variation in the ability of laboratory technician (intra) and technicians (inter) to determine the presence of malaria parasites in the same blood slide
- **Subject variation** – Eg. the response to a question may be affected by the subject's motivations and beliefs, and the place of interview.
- **Instrument and method variation** - some technology are more reliable than others.

Types of Error



Error Cont...

- Differential (Non-random)
 - If ascertainment failure of exposure or outcome variable is in one of the groups, it is differential (non-random) misclassification. We call this systematic error or bias
- Non-differential (Random)
 - (i.e. It is error influencing equally on the exposure and the outcome?)
 - If ascertainment failure is in both the groups, it is non-differential (random) misclassification

1. Random error

- ~ is an error occurring due to **chance**.
 - i.e. it is divergence from the true value due to chance.
- ~ Affects
 - the **precision** of a study
 - **Dilutes** the measure of association towards one side.
- ~ can never be eliminated,
 - But can be minimized by
 - **careful measurement** of exposure and outcome status
 - increasing **sample size** of study subjects.



Random Error

Arise due to;

**Biological (Individual)
variation**

Sampling error

**Measurement
(instrumental) error**

2. Bias (systematic error)

- ~ is when results differ from the true value in a **systematic manner**.
 - ✓ It is **undesirable**
 - ✓ It can't be '**adjusted for**'
 - ✓ Direction of effect is **unpredictable** since bias is difficult to prove
- ~ arising from the design or implementation of analysis of a study
- ~ Effect
 - affects the **validity** of a study
 - increase or decrease the **strength** of association (i.e. underestimation or overestimation).

Bias cont'd...

Type of Bias;

-Selection Bias

- Information Bias

- Can be minimized by;
 - Choosing appropriate study design
 - Using strict and randomized sampling procedure
 - Following the activities in a protocol strictly.
 - Choosing and sticking to standardized questioner
 - Training and blinding interviewers; so on.

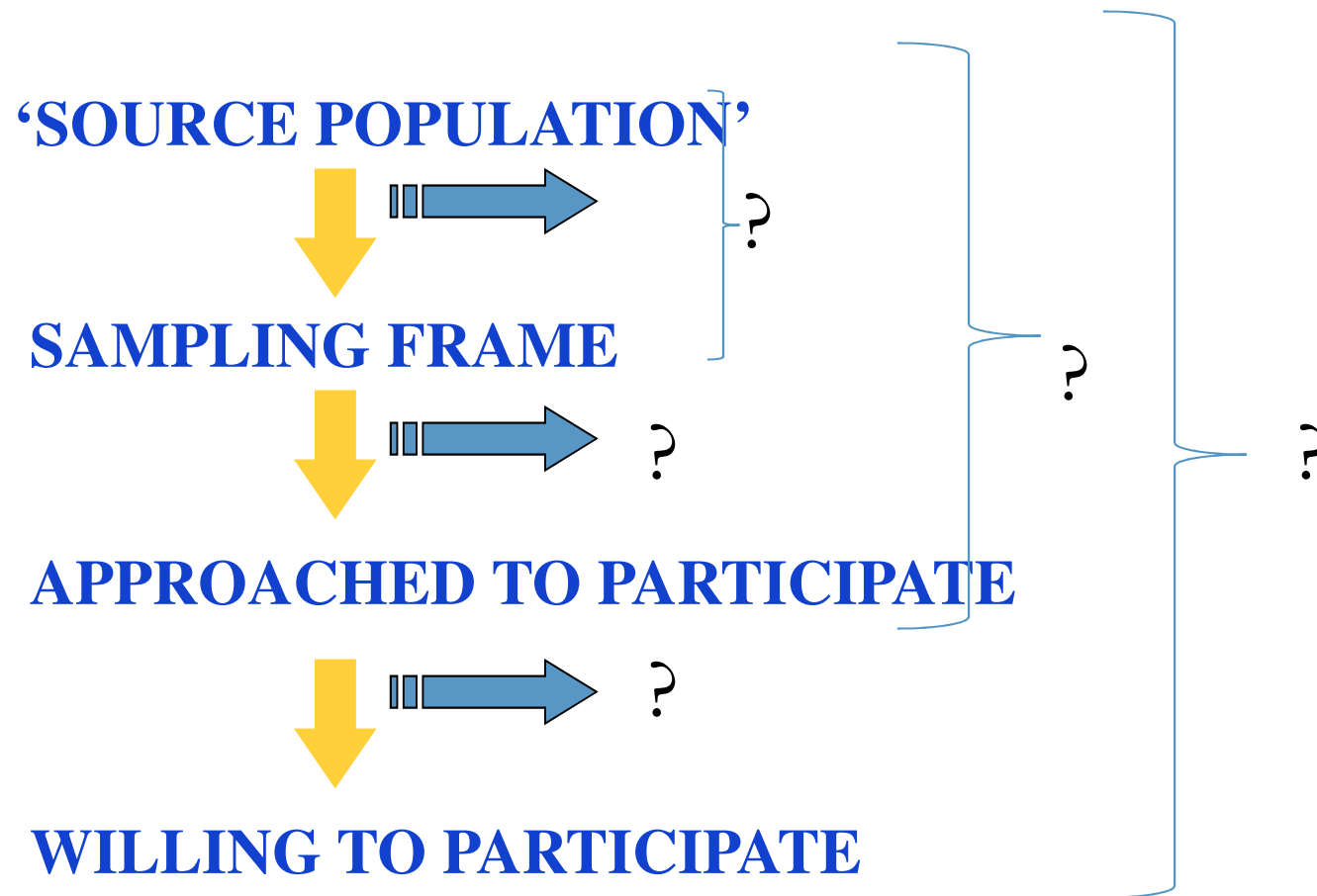
Selection Bias

- Refers to any error that arises in the process of identifying & selecting the study subjects.
 - Invitation
 - Acceptance
- Occurs when there is a systematic difference b/t characteristics of the people selected & those not selected.

Selection Bias

- **Invitational** (who gets invited into the study?)
 - Healthy worker bias ; Berkson's bias (differential hospital admission criteria, i.e. Hospitalization rates for case & control is different)
- **Acceptance** (who accepted the invitation?)
 - Loss to follow-up; Non-response bias
 - Volunteer/Compliance bias
 - Exclusion bias: different criteria for cases & control

‘Selection’



- WHO IS MISSING?
- DOES IT MATTER?

Selection bias is introduced;

- When selection of subjects depend on the **interest** of the **data collector/ investigator**
 - Study subjects does **not represent** the cases originated in the population
 - selection of study subjects follows **different criteria**, (eg. cases/controls)
- Selection bias is introduced in any of study designs but **more in case-control** and **surveys**

Summary: Minimizing Selection Bias

- Clear case definition:
- Give potential participants adequate explanation about the study
- Sensitize the community to the value of the research
- Appoint sensitive but persistent data collectors
- Stick in making appointments
- Use simple, concise instruments
- Feed back results
- *Record residence details*

Information Bias

- **Information Bias:** refers to any systematic error introduced during **measurement** of information (**assessing & measuring of information**) on either exposure or outcome variables
- In analytical studies, usually one factor is known and another is measured

E.g.

- In case control studies, the 'outcome' is known and the 'exposure' is measured
- In cohort studies, the exposure is known and the outcome is measured

Types of Information Bias

1. Interviewer bias
2. Recall bias
3. Observer bias
4. Lose to follow up

Minimizing information Bias

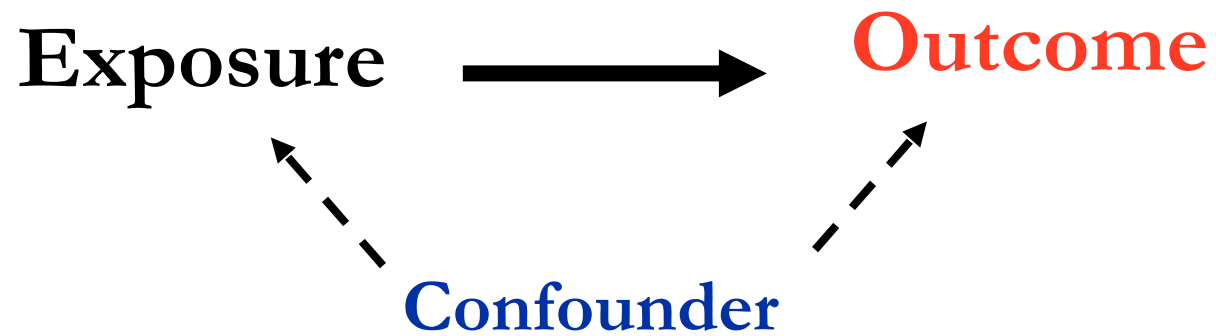
- Choose appropriate study **design**
 - Use the same method of assessment
- Use strict and randomized sampling **procedure**
- Follow strictly the **activities** in a **protocol**,.
- Use “Double-blinding” procedure
- Choose "hard" (objective) rather than “soft” (subjective) outcomes
- Choose and stick to **standardized** questioner & instrument
 - Use closed-ended questions whenever possible
 - Collect data on "dummy" variable
- **Train** and **blind** your interviewers; and so on

Confounder

Confounder

- **Disproportionate** existence of a factor b/t exposed & non exposed study subjects
 - i.e. a **risk factor** exists in a higher magnitude in one of the comparison group
- Measured effect of an exposure is distorted because of association of the exposure with other factor (3rd variable or confounder) that influences the outcome.

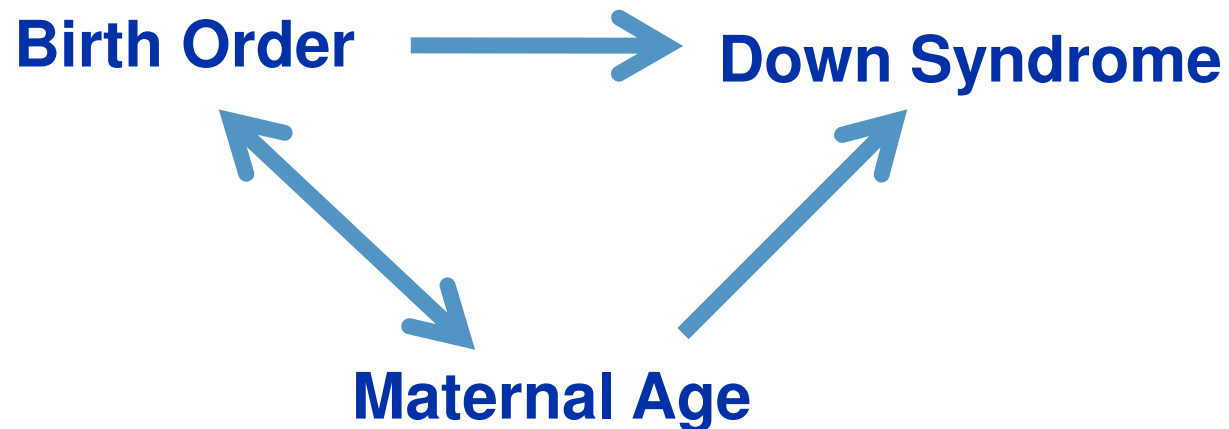
Confounder cont'd...



A confounding variable must satisfy four conditions

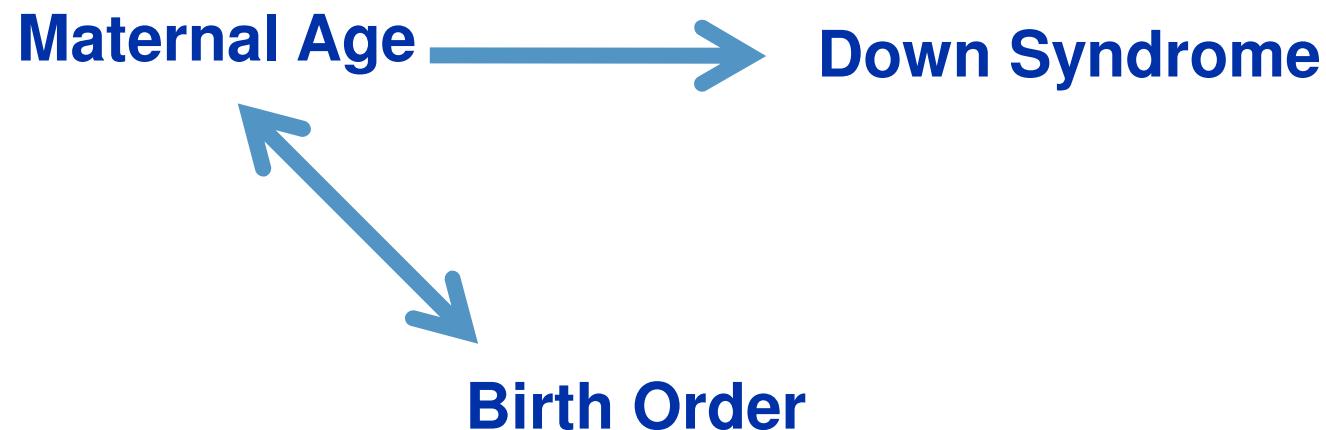
1. It should be associated with both the **exposure** and **outcome** variables
2. It must not lie on the **causal pathway** b/t exposure & disease
3. must be a **risk factor** for the disease or outcome (in the absence of the exposure of interest)
4. must not be affected by the exposure or the outcome i.e. should not be a consequence of either the exposure or disease).

Confounding



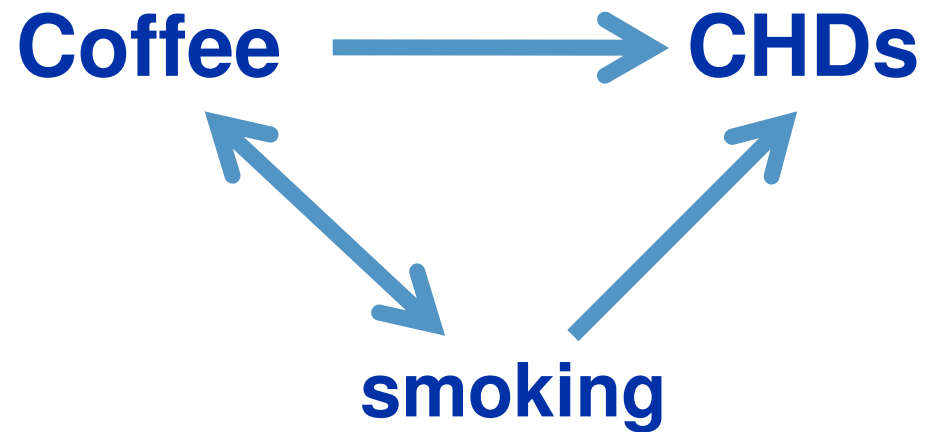
maternal age is correlated with birth order and a risk factor for Down Syndrome even if birth order is low

Confounding ?



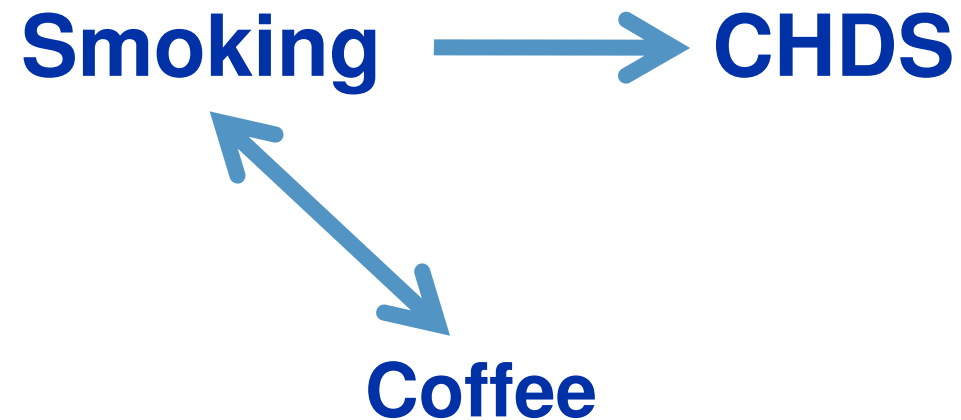
Birth order is correlated with maternal age but not a risk factor for Down Syndrome in younger mother

Confounding



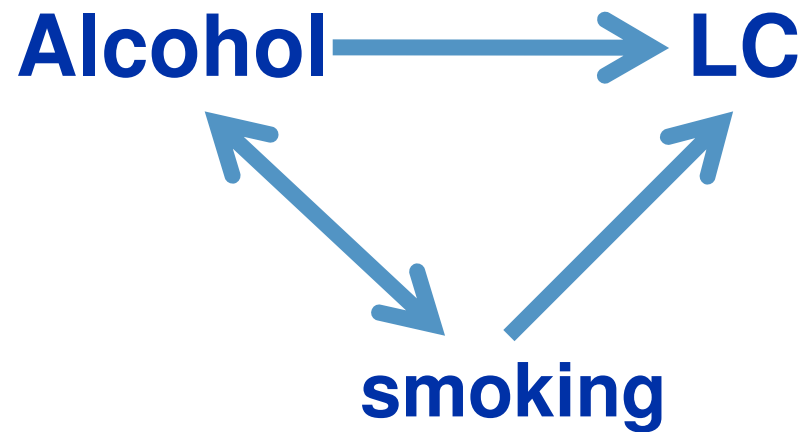
Smoking is correlated with coffee drinking and a risk factor even for those who do not drink coffee

Confounding?



**Coffee drinking may be correlated with Smoking but is
not risk factor in non smokers**

Confounding



Smoking is correlated with Alcohol and a risk factor for LC even for those who do not drink Alcohol

Type of Confounding

Two forms

- **Positive confounding** = Risk ratio or risk difference is **increased** from the true value by effect of the confounding variable
- **Negative confounding** = Risk difference or Risk Ratio are brought closer to the **null** by the effect of the confounding variable

Effects of confounding

A) Change the apparent **direction** of an effect

- May give false positive result,
- May give false negative result, i.e. **Mask** an underlying true association
- **Reverse** the actual direction of the association.

B) Lead to **overestimation** or **underestimation** of an effect, depending on the direction of the associations that the confounding factor has with exposure and disease.

How to minimize introduction of confounding

In the Study Design:

- Randomization
- Restriction
- Matching

In the Analysis:

- Stratification
- Multivariable Adjustment
- Propensity Scores

Chance

Chance

An association between exposure and outcome may occur by chance alone.

Consider chance at two stages

1. Designing phase

 Increase the sample size (the Power)

2. Analysis phase (Statistics)

- A. Hypothesis testing (test of **statistical significance**)

 P-value's role, (Chi square, Students-t-test, F-test, etc.

- B. Determination of **confidence interval**

Meaning of a Finding

- A finding of
 - “**no significant difference**” in a small study tells us nothing [because of sampling variation]
 - a “**significant difference**” in a small study may also tell nothing & may not be replicable
 - **no significant difference** in a large study tells us exposure & outcomes have no association
 - a **significant difference** in a large study reveals a true difference, but the findings may not be clinically important

Cont...

- Finding a significant association is not the **last** for conclusion
- Lacking to find a significant association is also **not the last** for conclusion
- Therefore, how do we reach in to a **CONCLUSION** about **causation** ?

Judgment of causation



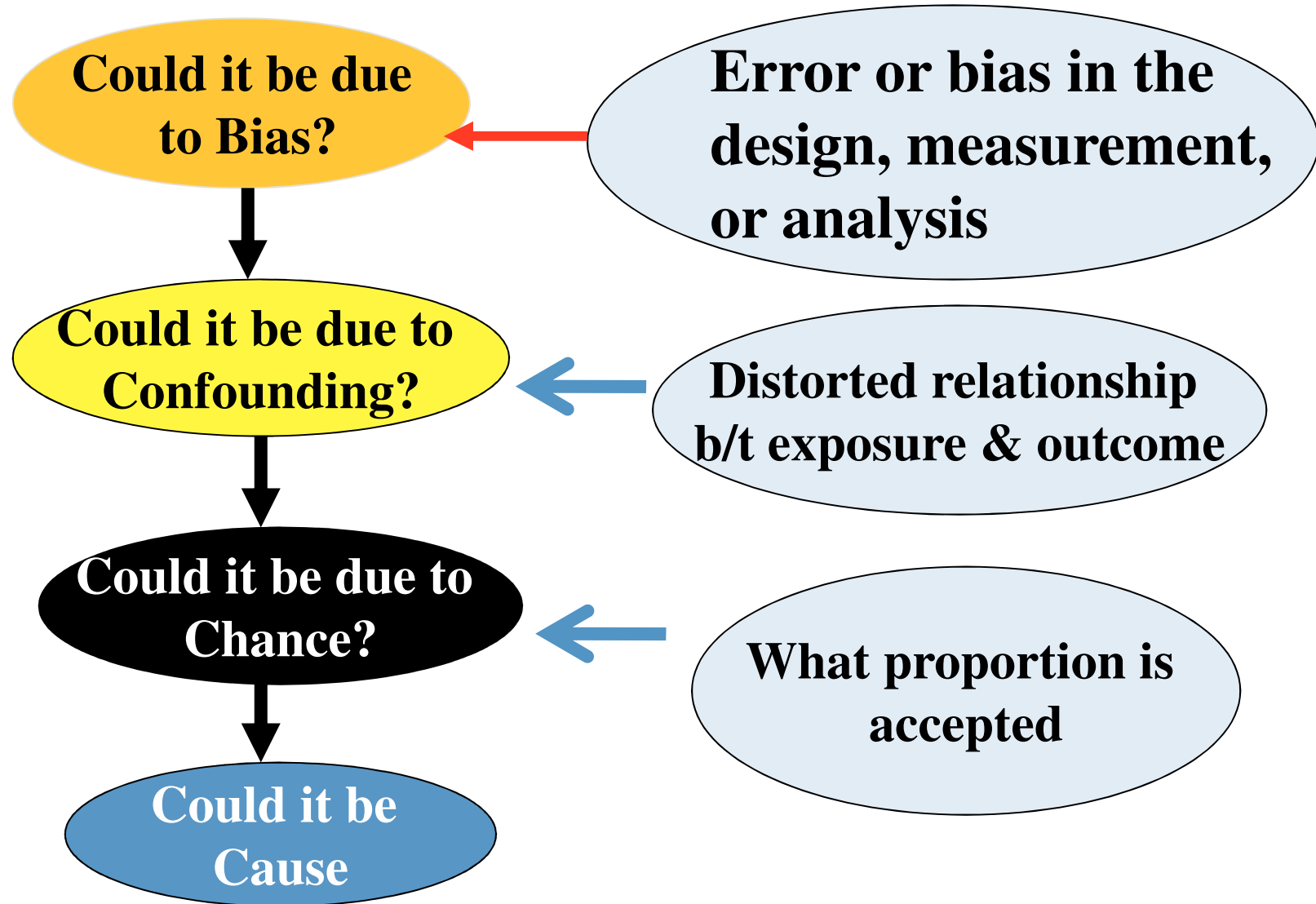
Establishing a Causal Association

- Once we found that chance, bias and confounding are all determined to be unlikely, then we can conclude that a **valid statistical association exists**.
- We should then apply for judgment of causality
- In the absence of experimental confirmation, the **'Bradford Hill' framework** is often applied.

▪

Establishing a Causal Association

Observed Association



The Bradford Hill Framework

- It is the statement of epidemiological criteria of a causal association formulated in 1965 by Austin Bradford Hill (1897-1991)
 1. Strength of association
 2. Consistency of relationship
 3. Specificity of association
 4. Temporal relationship
 5. Biologic gradient: Dose-response relationship
 6. Biological plausibility: explainable within the existing knowledge
 7. Reversibility (experimental): supported by experiment

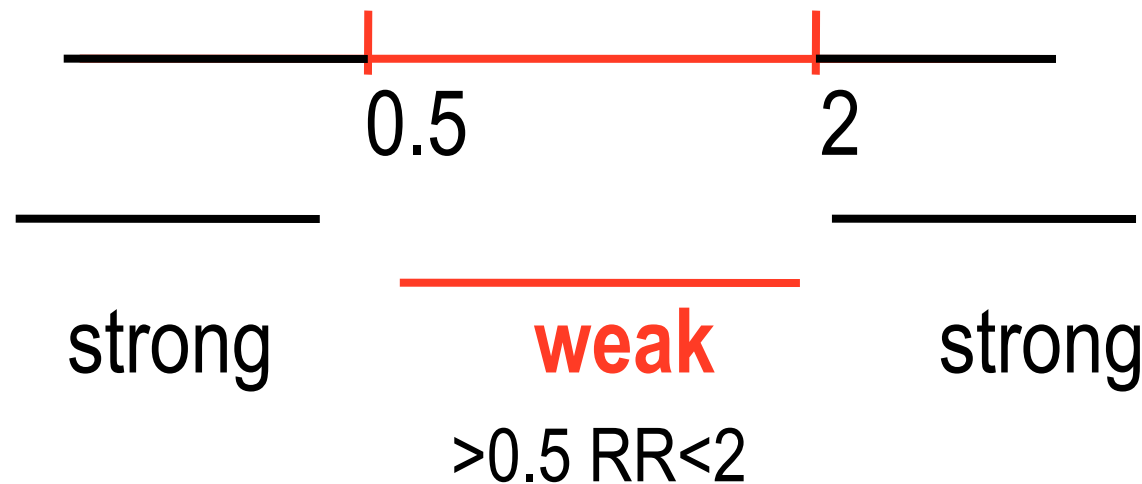
1. Strength of the Association

The stronger the association the more likely that it is a causal.

strong --- The more it is far from unity.

---RR ≥ 2 Weak < 2

---RR < 0.5 weak > 0.5



2. Consistency of Relationship

- The same association should be demonstrated by other studies both with different methods, settings and different investigators.
- That is why we compare our findings with other studies.
- Special methods of combining of a number of well designed studies exist, **Meta Analysis**.

3. Specificity of the association

Single exposure  Single disease

Plasmodium Sp.  **Malaria**

HIV  **AIDS**

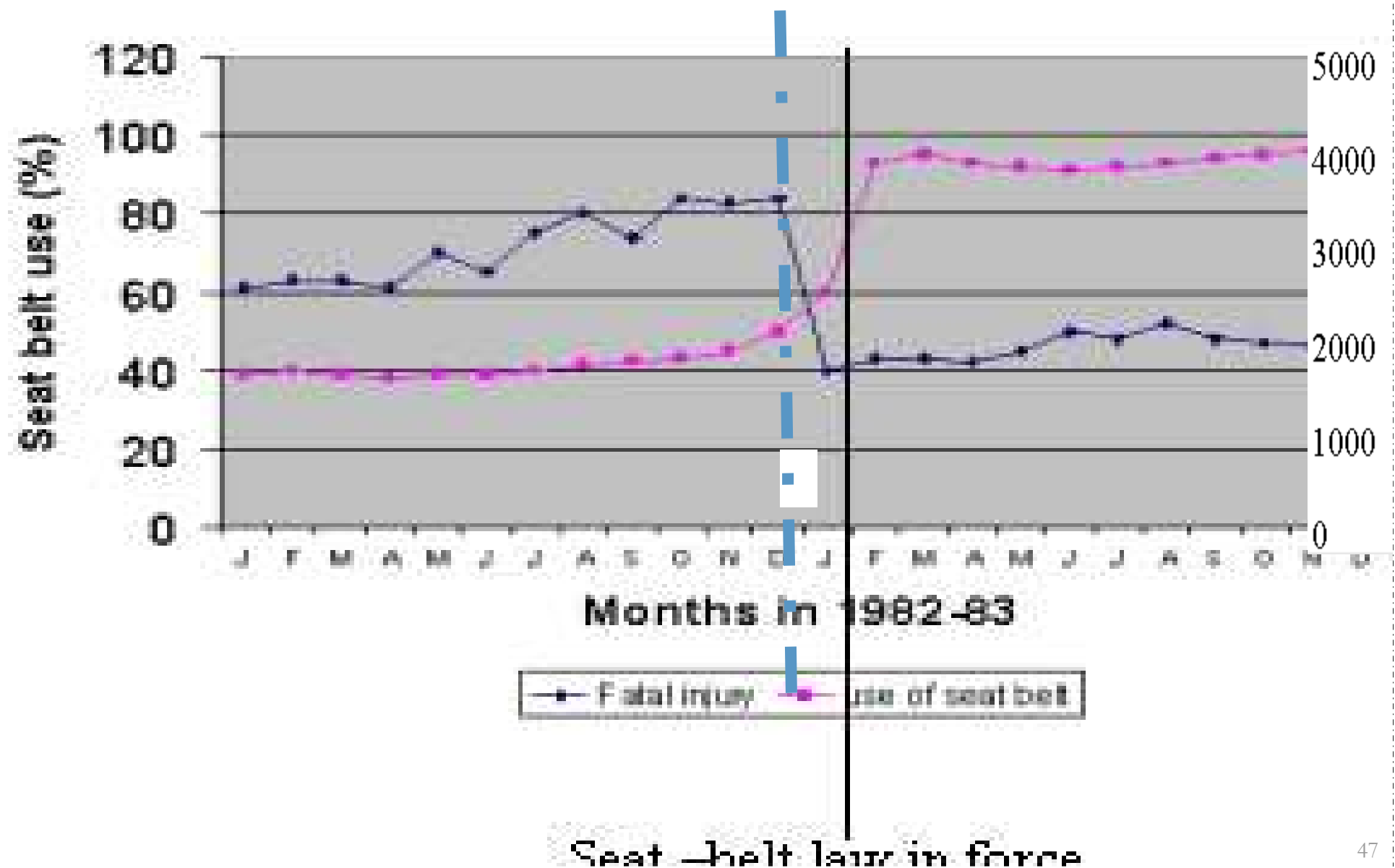
- This works more to most living organisms as causes.

4. Temporal relationship

- It is crucial that the cause must precede the outcome
- The study should show that exposure be observed first and outcome next
- This is usually problematic in cross-sectional and case-control designs.



Fig. 5. 4 Frequency of seat-belt use and injury occurrence in the United Kingdom. 1982-83



5. Dose response relationship

- The risk of disease increases with increasing exposure intensity.

E.g. death due to ischemic heart disease among male doctors above the age of 45 years followed in cohort

	Death rate per 100,000 pers-yr	Risk Ratio
Non-smokers	7	1.00
1-14 cigarettes/ day	46	6.57
15-24 cigarettes/ day	61	9.71
25+ cigarettes/ day	104	14.86

6. Biological plausibility

- Hypothesis should be coherent with what is known about the disease; both biologically and using laboratory.
- Knowledge about physiology, biology and pathology should support the cause-effect relationship
- If a finding doesn't go with what is known, then it is losing biological plausibility

7. Reversibility

- Removal of a possible cause results in a reduced disease risk

Eg. -Cessation of cigarette smoking is associated with reduction in risk of Lung cancer relative to those who continue.

-If the cause leads to rapid irreversible changes (as in HIV infection), then reversibility cannot be a condition for causality.

8. Study design

- It is most important to consider the design.

Type of study	Ability to give evidence
Randomized controlled trial	Most Stronger
Cohort study	Stronger
Case-control study	Moderate
Cross-sectional studies	Weak
Ecological studies	More weaker

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Tips in Judging the evidence

- There are no completely reliable criteria for determining whether an association is causal or not.
- In judging the different aspects of causation,
The correct temporal relationship is essential,
- Once this has been found, weight should be given to
 - Plausibility,
 - Consistency,
 - dose-response relationship and
 - Strength of the association

Sources

- Modern Epidemiology. Rothman & Greenland. LW&W, 1998.
- Principles of Epidemiology – Lecture Notes. Yemane Berhane, 2005.
- Epidemiology in Medical Practice. Barker & Rose. Churchill Livingstone, 1990.
- Use of acetaminophen and the risk of self-reported allergic symptoms and skin sensitization in Butajira, Ethiopia. Davey G, Berhane Y, Duncan P, Aref-Adib G, Britton J, Venn A. J Allergy Clin Immunol 2005;116:863-6.

The End!

